

# Chapter 2

## Series and Sequences

### 2.1 Infinite Series

#### Objectives

After completing this section, you should be able to:

- Define an infinite sequence, an infinite series, and the sequence of partial sums, and decide convergence of a series from its partial sums.
- Recognize a geometric series, and determine whether it converges and what it converges to.
- Apply the  $n$ th Term Test for divergence, and understand what the test can and cannot conclude.
- Combine convergent series using the sum, difference, and constant multiple rules, and evaluate telescoping series from their partial sums.

What does it mean to add infinitely many numbers together? In this section we answer that question by defining an infinite series and its sequence of partial sums, so that convergence of a series becomes a statement about a limit we already know how to study. We then put the definition to work on the geometric series, whose behavior we can settle completely, and on the  $n$ th Term Test, which is our first quick check for divergence.

*These are skeletal notes: the definitions, theorems, and problem statements are given, and blank boxes are left wherever a derivation, a solution, or a sketch belongs, so that you can fill them in by hand.*

#### 2.1.1 Definitions

**Definition 2.1.1 Sequence.** A **sequence** is a list of numbers,  $a_1, a_2, a_3, \dots$ . An **infinite sequence** of numbers is a function whose domain is the set of positive integers. ◆

**Example 2.1.2 A Sequence of Halves.** The numbers  $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$  form an infinite sequence; its  $n$ th term is  $a_n = \left(\frac{1}{2}\right)^n$ . ▲

**Definition 2.1.3 Infinite Series.** The sum of the numbers in an infinite

sequence  $\{a_n\}$ , i.e.  $a_1 + a_2 + a_3 + \cdots = \sum_{n=1}^{\infty} a_n$ , is called an **infinite series**.

Here  $a_n$  is the  $n$ th term of the series. ◆

**Example 2.1.4 An Infinite Series with a Finite Sum.** Infinite sequences can have finite sums. Consider the sum of the sequence from [Example 2.1.2](#), i.e.  $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$ .

It is most convenient to evaluate the result of this sum geometrically. Draw a square of side one in the space below. Shade half of it, then half of what is left, then half of what is left after that, and keep going. Label the pieces  $\frac{1}{2}$ ,  $\frac{1}{4}$ ,  $\frac{1}{8}$ ,  $\frac{1}{16}$ . What is the total shaded area, and therefore what is the sum?

*draw the unit square and its subdivisions here*



### 2.1.2 Partial Sums

Consider again the infinite sequence  $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$ . Let us denote the sum of the first  $n$  terms in this sequence by  $s_n$ , which are known as **partial sums**. Can we find a pattern in the sequence of partial sums?

Write out  $s_1$ ,  $s_2$ , and  $s_3$ , each one both as a fraction and in the form  $1 - \square$ .

Then guess a closed formula for  $s_n = \sum_{k=1}^n \left(\frac{1}{2}\right)^k$ .

$$s_1 =$$

$$s_2 =$$

$$s_3 =$$

$$\vdots$$

$$s_n =$$

Now use your formula for  $s_n$  to compute the infinite series, by taking the limit of the partial sum as  $n \rightarrow \infty$ . Compare the answer with the one you found geometrically from the square.

*workspace*

In the example above, the sequence of partial sums converged to a number. In general, if the sequence of the partial sums converges to a number, we say that the series **converges**, otherwise we say that the series **diverges**.

### 2.1.3 Geometric Series

An important example of infinite series is the geometric series. The **geometric series** is of the form

$$a + ar + ar^2 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1}, \quad (2.1.1)$$

where  $a, r$  are real numbers and  $a \neq 0$ . Note that we can re-write the series as  $\sum_{n=0}^{\infty} ar^n$ .

Let us compute the partial sum  $s_n$  for the geometric series, so that

$$s_n = a + ar + ar^2 + \cdots + ar^{n-1}. \quad (2.1.2)$$

Multiply (2.1.2) by  $r$ , subtract the result from (2.1.2), and solve for  $s_n$ . Almost every term should cancel.

$$r s_n =$$

$$s_n - r s_n =$$

$$s_n =$$

Next, take the limit of your partial sum as  $n \rightarrow \infty$ . Treat the cases  $|r| < 1$ ,  $|r| > 1$ , and  $r = 1$  separately, and in each case say what  $\lim_{n \rightarrow \infty} r^n$  does.

$$|r| < 1:$$

$$|r| > 1:$$

$$r = 1:$$

**Checkpoint 2.1.5 What Happens When  $r = -1$ ?** We have now handled  $|r| < 1$ ,  $|r| > 1$ , and  $r = 1$ , but one case is still missing:  $r = -1$ . Write out the partial sums  $s_1, s_2, s_3, s_4$  of the series  $a - a + a - a + \cdots = \sum_{n=1}^{\infty} a(-1)^{n-1}$  for  $a \neq 0$ . Does the sequence of partial sums approach a single number as  $n \rightarrow \infty$ ? What does that tell you about the series?

*Your answer and reasoning.*

*workspace*

**Hint.** The partial sums do not grow without bound here, the way they do

when  $r = 1$ . Look instead at whether they settle down to one value.

Putting all of the cases together, we can say that if  $|r| \geq 1$ , the geometric series is divergent.

**Summary.**

The geometric series (2.1.1) converges to  $\frac{a}{1-r}$  if  $|r| < 1$ , i.e.

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \quad |r| < 1, \quad (2.1.3)$$

and diverges if  $|r| \geq 1$ .

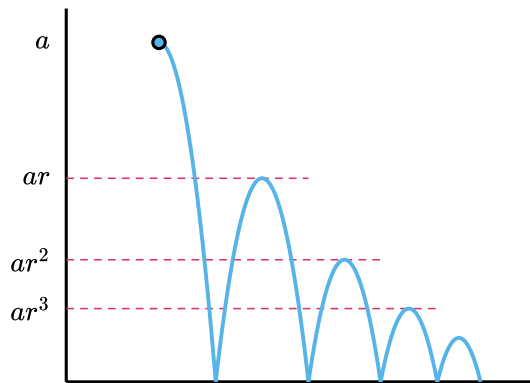
Next, as an application of the geometric series, we will go through the following example, which is from our textbook.

**Example 2.1.6 A bouncing ball.** You drop a ball from  $a$  meters above a flat surface. Each time the ball hits the surface after falling a distance  $h$ , it rebounds a distance  $rh$ , where  $r$  is positive but less than 1. Find the total distance the ball travels up and down.

*Solution.* (Use Figure 2.1.7 to write the total distance as an infinite series, then match it against (2.1.3). Watch the first drop: it is travelled only once.)

*workspace*





**Figure 2.1.7** The ball falls a distance  $a$ , then rises and falls a distance  $ar$ , then  $ar^2$ , and so on, so the total distance travelled is  $a + 2ar + 2ar^2 + \dots$ .

### 2.1.4 The $n$ th Term Test

**Theorem 2.1.8** If  $\sum_{n=1}^{\infty} a_n$  converges, then  $\lim_{n \rightarrow \infty} a_n = 0$ .

**Important note:** If  $\lim_{n \rightarrow \infty} a_n = 0$ , we *cannot* conclude that  $\sum_{n=1}^{\infty} a_n$  converges. See parts D and E in the example below.

The following test is a consequence of the above theorem.

#### The $n$ th Term Test.

If  $\lim_{n \rightarrow \infty} a_n$  does not exist or  $\lim_{n \rightarrow \infty} a_n \neq 0$ , then  $\sum_{n=1}^{\infty} a_n$  diverges.

**Example 2.1.9 Testing series for convergence.** Determine whether the series is convergent or divergent. If it is convergent, find its sum.

A.  $\sum_{n=1}^{\infty} n$

B.  $\sum_{n=1}^{\infty} (-1)^n$

C.  $\sum_{n=1}^{\infty} \frac{n^3 + 1}{n^2 + n}$

D.  $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

E.  $\frac{4}{9} - \frac{8}{27} + \frac{16}{81} - \frac{32}{243} + \dots$

*Solution.*

A.

B.

C.

D.

E.



### 2.1.5 Combining Series

**Theorem 2.1.10** If  $\sum a_n = A$  and  $\sum b_n = B$  are convergent series, then

1. Sum Rule:  $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. Difference Rule:  $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. Constant Multiple Rule:  $\sum ka_n = k \sum a_n = kA$  (Any number  $k$ ).

**Example 2.1.11 Using the difference rule.** Evaluate  $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} - \frac{1}{2^n}$ .

*Solution.* (You have already summed both of these series in this section: one in [Subsection 2.1.2](#) and the other in part D of [Example 2.1.9](#).)

workspace



**Example 2.1.12 A telescoping series with logarithms.** Evaluate  $\sum_{k=0}^{\infty} \ln\left(\frac{k+2}{k+1}\right)$ , if it converges and if it diverges, show that it does.

*Solution.*

workspace

